

CII-SR Kaizen Circle Competition 2018

A Kaizen Presentation By

(EVEREST INDUSTRIES LIMITED)

On: 26th July-18

Venue : Hotel Westin, Chennai

Company Overview:

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing, ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of designs, drawings and implementation for every project.

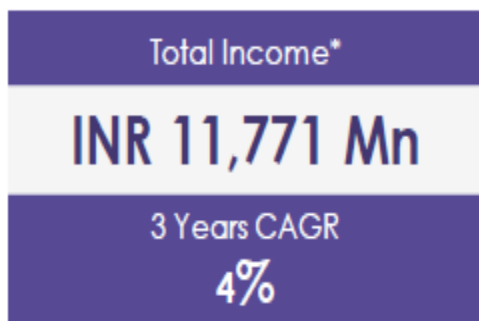
FY17 Business Mix (Revenue Share %):

- **Building products (63%)** – includes fibre cement roofing sheets, fibre cement boards, solid wall panels.
- **Steel buildings (37%)** – offers customised building solutions like Pre-Engineered Steel Buildings, Smart Steel Buildings, metal roofing sheets and cladding.

Manufacturing and Distribution Networks:

- 6 Building Products plants and 3 Steel Building Plants.
- 40 Sales Depot, over 6,000 Dealer Outlets, serving over 600 cities & 100,000 villages.
- Export to over 35 countries (Green solutions – Fibre cement boards).

FY17 Financial Snapshot (Consolidated):



*Total income includes other income

Everest range of products includes



Fibre Cement Roofing
Sheets & Accessories



Hi-Tech Roofing Sheets



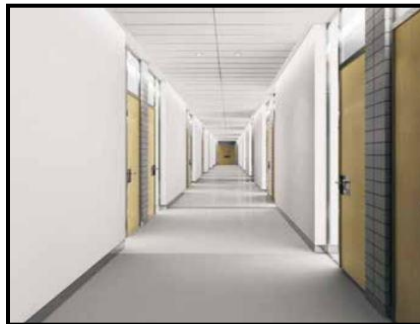
Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings



Reliance
Industries Limited



Walmart
Save money. Live better.

IFB

ACC



Mahindra

HYATT
CAPITAL
GATE
ABU DHABI

Pidilite

TATA

Whirlpool
CORPORATION

RELIANCE



Marriott
HOTELS & RESORTS

ITC
ITC Limited

BRITANNIA
Eat Healthy. Think Better

DLFA

DLFA
BUILDING INDIA

एन वी सी सी
NBCC
A Navratna CPSE

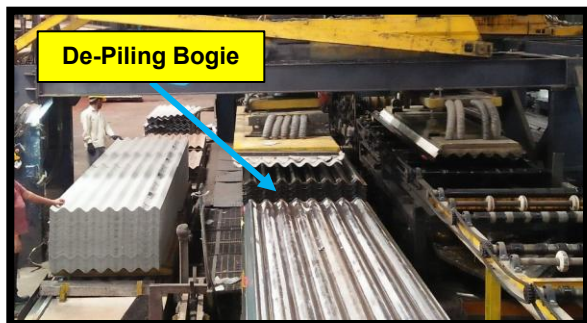
Cipla

Coca-Cola

VIDEOCON

Kaizen Theme: To improve productivity by recovering Green sheets during De-piling bogie changeover to avoid Green sheet wastage.

Implemented Area: Sheeting Machine -1 De-Piling Area at Podanur Works, Coimbatore



Cross Functional Team:



Mr. Y. Kesavaraj
Line Incharge



Mr. N. Karthikeyan
Shift Incharge



Mr. B.N.V. Manikanta
Shift Incharge



Mr. M. Vijay Kumar
Electrical Maint.



Mr. N. Anil Kumar
QS & BE



Team Leader
Mr. S. Rajesh Kumar
(Maint.)



Mr. Narayanasami
Board Operator



Mr. Nagraj
Board operator

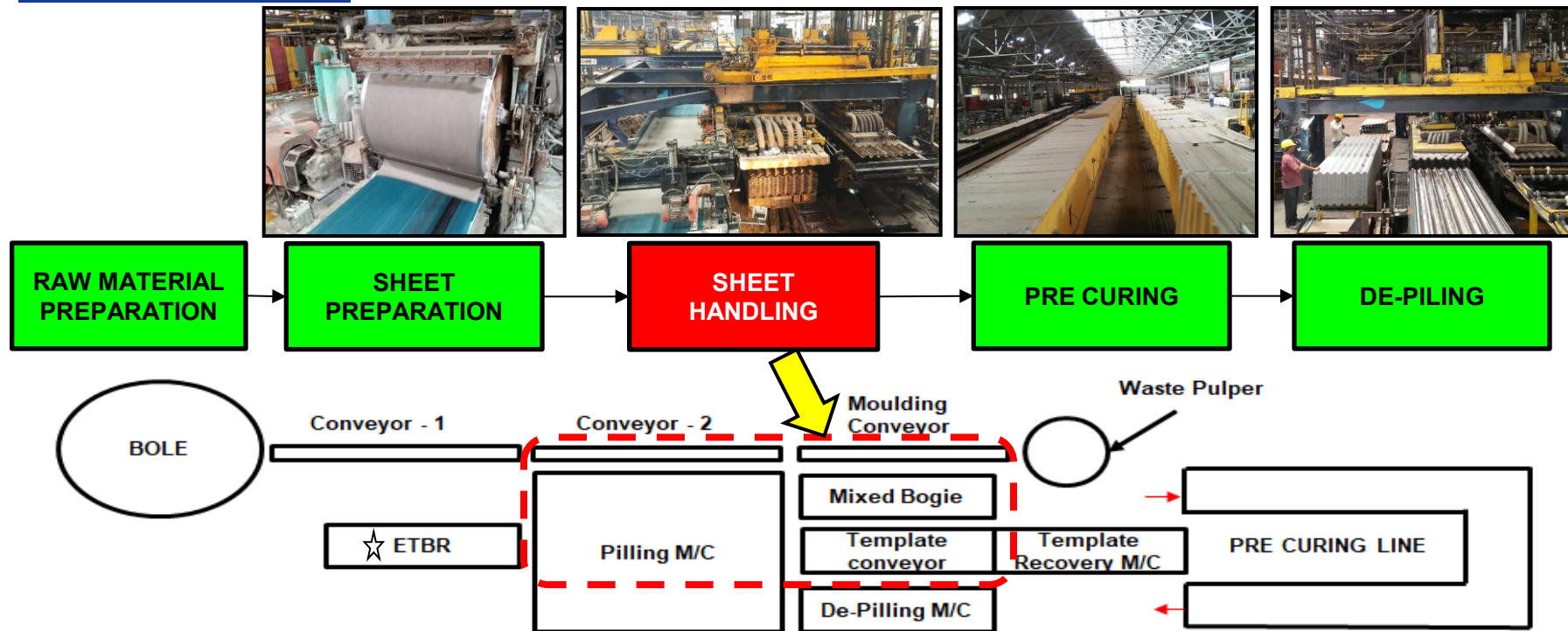


Mr. Prasadhan
Fitter



Mr. Pillai
Fabricator

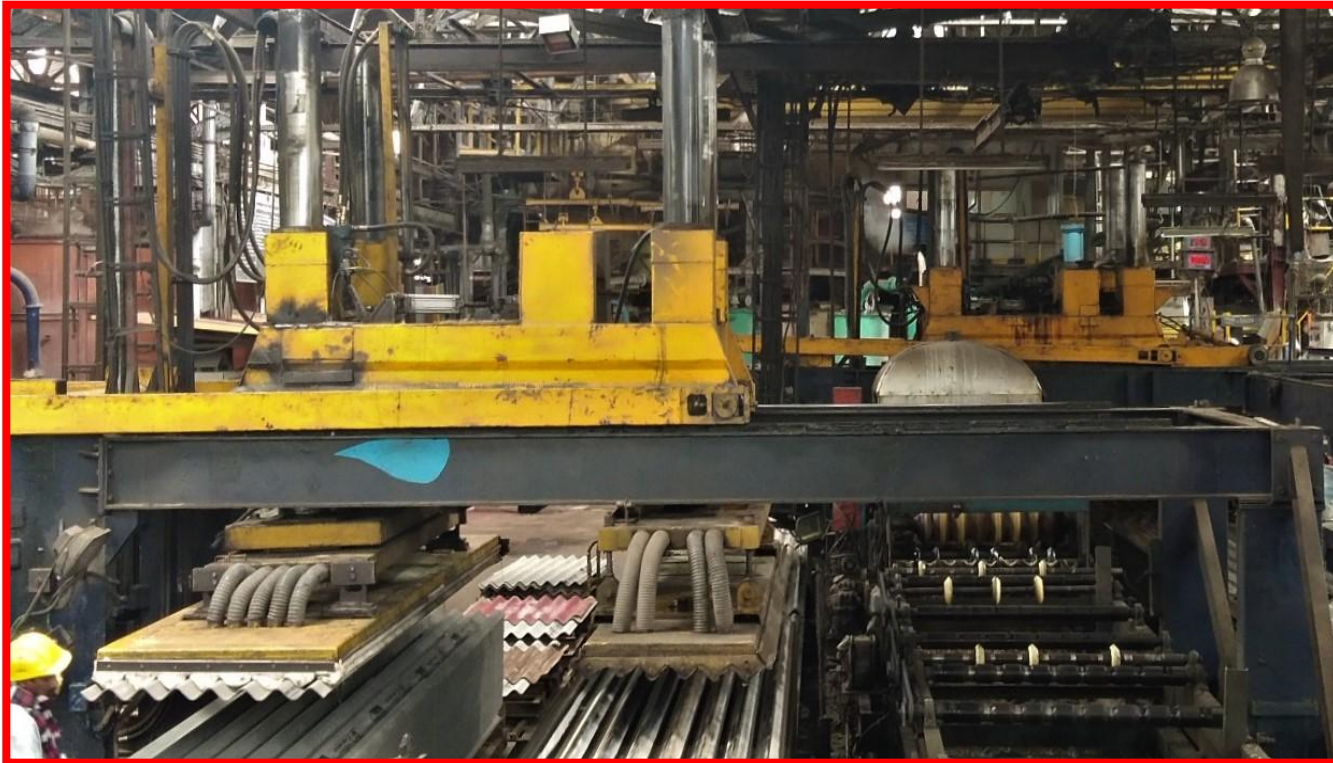
Process Flow:



Due to Non-availability of Metal Templates for Piling Mixed bogie during De-piling Bogie changeover there is Loss of Green sheets (2 sheets / Cycle).

- Potential loss of approx. 3.02%
- Approximate Loss = 9.82 MT/day ~ INR 98,200 Revenue loss/day.

☆ TBR – Transborder Rail



- There was no additional template roller conveyor to store templates during Piling bogie changeover. And to supply stored templates for Mixed bogie piling during De-piling bogie changeover.

EVEREST Root Cause Analysis & Idea Generation

Problem: During De-piling bogie changeover there is loss of Green sheets due to Non-availability of templates.

Why 1 : There are no Templates to provide for Mixed bogie piling during De-piling bogie changeover.

Why 2 : It takes some time for De-piling bogie changeover and at that time there is no mechanism for providing templates.

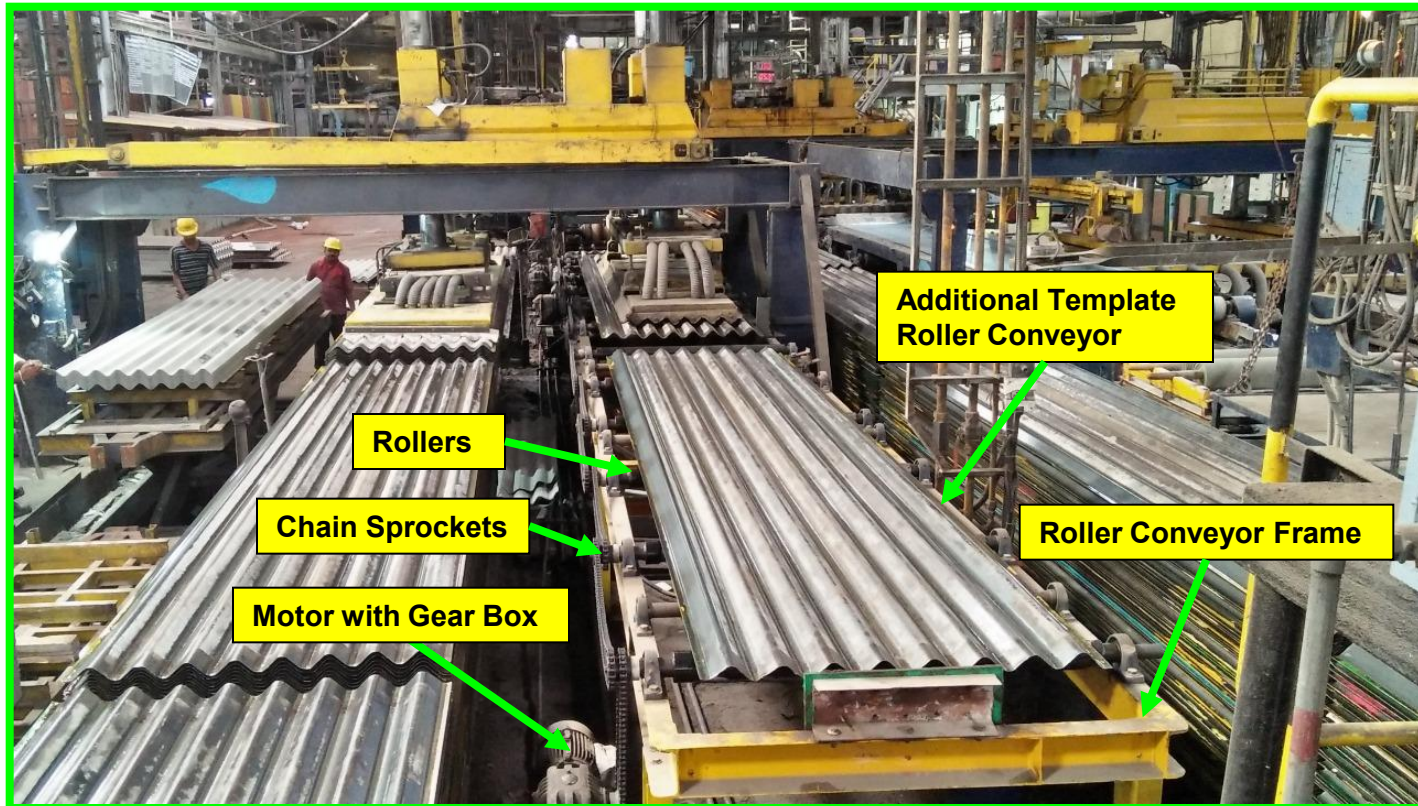
Why 3 : Machine / Design Constraint

Root Cause: No mechanism to supply Metal Templates during De-piling bogie changeover.

Idea Generation: On line supply of additional templates.

Action: Installation of additional template roller conveyor to store 2 Templates during Mixed bogie changeover & to provide stored templates to Mixed bogie piling during De-piling bogie changeover.

SN	Counter Measure Activity
01	Space arranged for extending the template roller conveyor system.
02	Installation of roller conveyor frame.
03	Installation of rollers, Gearbox, Motors & coupled them with chain sprockets.
04	Providing Pneumatic cylinder arrangement to lift the template on the roller conveyor to store another template on the roller conveyor.
05	Giving control circuit at De-piling Board & taken safety measures.



- Installed additional template roller conveyor to store templates during piling bogie changeover and to provide stored templates for piling mixed bogie during De-piling bogie changeover.

Quantitative : Potential Cost Saving of 1.12 Cr./Annum.

Productivity : Average monthly productivity increased from 278 MT to 288 MT.

Quality : Quality of sheets improved due to reduction in recycled waste by 16.32%.

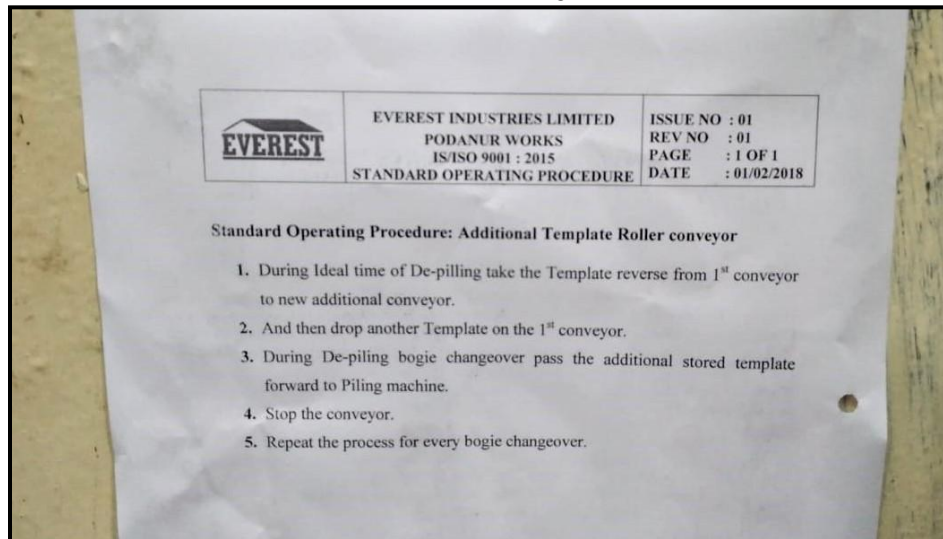
Cost : Potential Cost Savings = 1.12 Cr./Annum

Safety : The countermeasure is being controlled by PLC. Hence the Idea implemented is safe.

Morale : This Kaizen has been awarded as the Best Kaizen of Quarter 3, 2017-18 in Inter-works competition.



- Training on Standard Operating Procedures are given to all the operators and maintenance people to operate the new system.



Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Scope of Horizontal Deployment in across Everest :-

Factory	Target Date	Status
1.Bhagwanpur Factory, Roorkee	31.07.2018	Under Implementation
2.Kolkata Factory, Kolkata	30.07.2018	Under Implementation
3.Kymore Factory, Madhya Pradesh	15.08.2018	Under Implementation
4.Podanur Factory, Coimbatore	31.03.2018	Implemented in both the lines
5.Somnathpur Factory, Odisha	30.07.2018	Under Implementation

Kaizen Sheet

Company

MM/YY

S.No

Productivity

Quality

Delivery

Safety

Material

Energy

Maintenance

Everest Industries Ltd

11/17

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Kaizen Title: To improve productivity by recovering Green sheets during De-piling bogie changeover to avoid Green sheet wastage.

Implemented Area:

Sheeting machine-1 Shopfloor

Problem/Present Status:

Loss of Green sheets(2 sheets/cycle) during De-piling bogie changeover. This is a potential loss of approx. 3.02 % which if avoided can lead to productivity improvement. The loss is approx. 9.82 MT/Day i.e. INR 98,200 revenue loss/Day.

Before Improvement:



Implemented by:

S. Rajesh(TL), Y. Kesavaraj, M. Vijay, Karthikeyan, Manikanta, N.Anil, Pillai, Prasadhan, Narayanasami, Nagaraj

Result/Benefit:

(a) Qualitative

Quality of sheets improved due to reduction in recycled waste by 16.32%.

Root Cause Identification:

Problem

•During De-Piling bogie changeover there is loss of green sheets due to non availability of Templates.

Why 1

•There are no Templates to provide for Mixed bogie piling during De-Piling bogie changeover.

Why 2

•It takes some time for De-Piling bogie changeover and at that time there is no mechanism for providing templates.

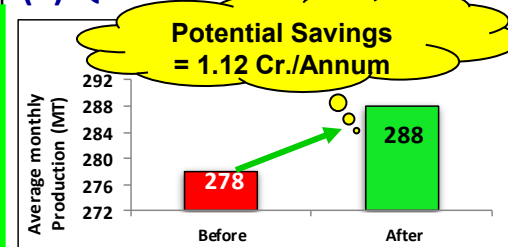
Why 3

•Machine/Design Constraint

After Improvement:



(b) Quantitative



Standardization:

Preventive maintenance as per schedule.

Root cause: No mechanism to supply Templates during De-Piling bogie changeover.

Idea to eliminate root cause: On line supply of additional templates.

Counter-measure:

Installed additional template roller conveyor to store template during Mixed bogie changeover & to provide stored template to mixed bogie piling during De-piling bogie changeover. By this system there is no loss of Green sheets during De-piling bogie changeover.

Horizontal Deployment:

- 1) Bhagwanpur Unit - Under Implementation
- 2) Kolkata Unit – Under Implementation
- 3) Kymore Unit – Under Implementation
- 4) Podanur Unit – Implemented
- 5) Somnathpur Unit – Under Implementation